BJT operation and DC bias I

Cheat sheet · Day 06 · Thu 29 Oct · pack E4

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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BJT MODES (lecture table; the WS4 handout has sat/cut-off rows swapped)
  Active: EBJ fwd, CBJ rev | VBE=0.7, IC=bIB      | check IB>0, VCE>0.2
  Sat:    EBJ fwd, CBJ fwd | VBE=0.7, VCE=0.2     | check IB,IC>0, IC<bIB
  Cutoff: both rev         | IB=IC=IE=0           | check VBE<0.7
CURRENTS  IE=IC+IB   IC=bIB   IE=(b+1)IB   IC=aIE
  a=b/(b+1)   b=a/(1-a)   ICEO=ICBO/(1-a)=(b+1)ICBO
  b_dc = ICQ/IBQ at the Q-point   (b_ac = dIC/dIB, curve spacing)
BIAS (assume active, then CHECK)
  Fixed:     IB=(VCC-0.7)/RB                VCE=VCC-IC*RC
  Emitter:   IB=(VCC-0.7)/(RB+(b+1)RE)      VCE=VCC-IC*RC-IE*RE
  Divider exact: RTh=R1||R2, ETh=VCC*R2/(R1+R2)
                 IB=(ETh-0.7)/(RTh+(b+1)RE)
  Divider approx ONLY IF b*RE >= 10*R2  (write the test!)
                 VB=ETh, VE=VB-0.7, IC~IE=VE/RE, VCE~VCC-IC(RC+RE)
  Two supplies on the base -> superposition for VTh (PT Q7)
  Node voltages: VE=IE*RE, VC=VCC-IC*RC, VB=VE+0.7, VBC=VB-VC (<0 active)
LOAD LINE  VCE=VCC-IC(RC+RE):  (VCC,0) and (0, VCC/(RC+RE)=ICsat)
  Q = load line meets IB=IBQ curve; write the IC, VCE you read; b=ICQ/IBQ
  Design: VCC from VCE intercept, RC(+RE)=VCC/ICsat,
          RB=(VCC-0.7-IE*RE)/IBQ -> nearest standard value, quote % error
Negative VCE from an active calc => saturated, redo (pack E5)
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